

Host Identification — Wireshark Investigation Report

Network Forensics | Incident Response Practice

Date: 2026-06-13 | Source: Malware-Traffic-Analysis.net (2026-02-28 Exercise) | By: Jessica Sansaricq

1. Objective

This investigation focused on identifying key host attributes of a known-infected machine on a network segment, using a provided PCAP file from Malware-Traffic-Analysis.net (2026-02-28 — "Easy as 123" exercise). The primary goal was to extract the MAC address, hostname, Windows username, and full user name associated with the infected host at IP address 10.2.28.88.

Host identification is a critical early step in incident response. Mapping a suspicious IP address to a physical machine, a Windows domain account, and ultimately a real user allows security teams to scope the breach, isolate affected systems, review account activity, and initiate remediation — all with confidence that the investigation is targeting the correct endpoint.

2. Tools Used

Primary Analysis Tool: Wireshark 4.x — open-source packet capture analyzer

Exercise Source: Malware-Traffic-Analysis.net — 2026-02-28 Easy as 123 PCAP & answer key

Reference: Answer key published by Brad Duncan at malware-traffic-analysis.net

Wireshark filters applied during the investigation:

- `ip.addr == 10.2.28.88` — isolate all traffic to/from the infected host
- `nbns` — reveal NetBIOS Name Service registration packets for hostname discovery
- `eth.addr == 00:19:d1:b2:4d:ad` — confirm MAC address in Ethernet II headers
- `kerberos.CNameString` — extract the Windows username from Kerberos AS-REQ packets
- SAMR traffic review — locate full account details including display name

3. Known Infected Host

IP Address: 10.2.28.88

Confirmed By: Previous investigation — Wireshark filter `ip.addr == 10.2.28.88`

Applying the infected-host filter removed all unrelated traffic from the PCAP view and confirmed the host was actively communicating across multiple protocols including DNS, Kerberos, SMB, LDAP, and HTTP-related traffic. This multi-protocol activity is consistent with normal Windows domain workstation behaviour and provided a clean baseline for further attribute discovery.

4. MAC Address Findings

Filter Used: `eth.addr == 00:19:d1:b2:4d:ad`

Location: Ethernet II → Source Address field in packet details

MAC Address Found: 00:19:d1:b2:4d:ad

The Ethernet II layer of packets originating from 10.2.28.88 consistently showed the source MAC address 00:19:d1:b2:4d:ad. The MAC address uniquely identifies the network interface card (NIC) of the infected machine and is useful for correlating the IP address to a specific physical device in switch ARP tables and DHCP lease logs.

5. Hostname Findings

Protocol: NBNS — NetBIOS Name Service

Filter Used: `nbns`

Location: NBNS registration packet — Name field in packet details

Hostname Found: DESKTOP-TEYQ2NR

An NBNS name registration packet broadcast from 10.2.28.88 announced the Windows computer name DESKTOP-TEYQ2NR. NetBIOS name registration occurs when a Windows host boots and registers its name with the local network, making it a reliable early-stage signal for hostname identification without requiring access to Active Directory logs.

6. Username & Full Name Findings

6a. Windows Username (Kerberos)

Filter Used: kerberos.CNameString

Location: Kerberos → AS-REQ → CNameString field

Username Found: brolf

Filtering on the Kerberos CNameString field exposed authentication requests made by the infected host. The CNameString in a Kerberos AS-REQ (Authentication Service Request) contains the plaintext Windows username of the account seeking a Ticket-Granting Ticket (TGT). The username brolf was clearly visible in these packets, linking the infected machine to a specific domain account.

6b. Full Name (SAMR)

Method: SAMR (Security Account Manager Remote Protocol) traffic review

Location: SAMR decoded packet details → Full Name field

Full Name Found: Becka Rolf

After identifying the username brolf, SAMR traffic in the PCAP was reviewed for Windows account attribute details. SAMR is used by Windows to query and manage account information in Active Directory. The decoded SAMR packet revealed the display (full) name field as Becka Rolf, connecting the domain account brolf to a real individual — information that is essential for notifying affected users and performing account-level remediation during an incident response.

7. Answer Key Comparison

The table below compares every finding from this investigation against the official answer key published by Malware-Traffic-Analysis.net for the 2026-02-28 exercise.

Field	Value Found	Discovery Method / Filter	Answer Key
IP Address	10.2.28.88	Wireshark — infected host filter (ip.addr)	✓ Match
MAC Address	00:19:d1:b2:4d:ad	Wireshark — Ethernet II / eth.addr filter	✓ Match
Hostname	DESKTOP-TEYQ2NR	Wireshark — NBNS registration traffic (nbns)	✓ Match
Windows Username	brolf	Wireshark — Kerberos CNameString filter	✓ Match
Full Name	Becka Rolf	Wireshark — SAMR account details	✓ Match

All five findings matched the answer key exactly. No discrepancies were identified.

8. Analyst Reflection

Identifying the IP address and MAC address was straightforward — both appeared immediately after applying the host filter and inspecting Ethernet II headers, requiring no protocol-specific knowledge beyond basic Wireshark navigation. The hostname discovery via NBNS was equally quick once the protocol filter was applied, as the registration packet broadcasted the computer name in plaintext.

The username discovery through Kerberos CNameString was slightly more specialised, requiring familiarity with authentication protocols, but the Wireshark column filter made the value immediately visible. The most involved step was locating the full name in SAMR traffic, which required manually drilling into decoded packet fields rather than relying on a single filter — reinforcing that deeper investigation often demands understanding of Windows domain protocols beyond surface-level packet inspection.

This exercise strengthened the workflow of progressively layering Wireshark filters to move from raw IP traffic to specific protocol data, and highlighted how standard Windows authentication and name resolution mechanisms leave rich forensic artefacts in network captures.

End of Report — Host Identification | Wireshark Investigation